UNITS OF THE GLUED RING SPLIT THROUGH A VALUATION RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement sits in Situation situation-glue (the glueing setup for dimension-one rings).

Allowed standard prerequisites (editorial).

- For an element of a fraction field of a valuation ring, x or x^{-1} lies in the ring. Algebra, Lemma algebra-lemma-valuation-ring-x-or-x-inverse. The dichotomy.

Source reference. Original reference 09MY: sources/stacks-chow/tags.txt, source lines 12196–12196. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00IB: sources/stacks-chow/tags.txt, source lines 673–673. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 09MY assume that B is a valuation ring. Then for every unit u of A either $u \in R$ or $u^{-1} \in R$.*

Proof. Namely, if the image c of u in K is in B, then $u \in R$. Otherwise, $c^{-1} \in B$ (Algebra, Lemma 00IB) and $u^{-1} \in R$. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.